

KARTHIKEYAN SIVASUBRAMANIAN

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EDUCATION

Arizona State University (School of Computing and Augmented Intelligence (SCAI))

Aug 2024 - May 2026

Master of Science, Computer Science (GPA 4.0/4.0)

Tempe, AZ

Vellore Institute of Technology

Sep 2020 - May 2024

Bachelor of Technology, Electronics and Communication Engineering (GPA 8.82/10)

Chennai, Tamil Nadu

EXPERIENCE

School of Computing, Arizona State University

Aug 2025 - Present

Graduate Teaching Assistant (Grader)

Tempe, AZ

- Mentored and evaluated students in compiler design, lexical and syntax analysis, top-down parsing, runtime systems, and lambda calculus, providing structured feedback on lectures, projects, and assessments.

Voltech Group

July 2023 - Aug 2023

Embedded Systems Design Intern

Chennai, Tamil Nadu

- Built an IoT-based telemetry pipeline** for SMPS battery chargers using a **NodeMCU web server**, acquiring voltage and temperature via **ADCs** and enabling real-time operational monitoring.
- Extended the pipeline with **GRU/LSTM-based time-series forecasting** on voltage-current data, **achieving ~20% lower RMSE** over a naïve baseline and operationalized model inference via **AWS SageMaker, API Gateway, Lambda, and CloudWatch**.

Center for Advanced Data Science, Vellore Institute of Technology

Mar 2023 - Oct 2023

ML Research Intern

Chennai, Tamil Nadu

- Achieved 97% average accuracy** in multilingual signature classification across 4 languages, by designing a **two-stage Siamese network** with **MobileNet-V2**, trained using **TensorFlow tf.data (parallel map, prefetch)** and A100 GPU acceleration, **reducing training time by ~18%** through input-pipeline optimization.
- Attained 88% recall** in signature detection by implementing **YOLOv5/v7/v8 (DataLoader parallelism)** for region extraction and deploying **MobileNet-V2** via **AWS SageMaker endpoints**, with a lightweight detection pipeline running locally on a **Raspberry Pi 4**.

TECHNICAL SKILLS

Languages/Frameworks: C++, Python, Java, TypeScript, Tailwind, React.js, Node.js, Spring Boot, Django, FastAPI, Pydantic, Flask, Streamlit, Gradio, Scikit-learn, OpenCV, PyTorch, Tensorflow, Keras, Apache Spark

OS/Tools/Databases: Linux, Git, Docker, Kubernetes, Kafka, Airflow, REST APIs, OAuth, Terraform, AWS (ECR, EC2, S3, RDS, Lambda, Beanstalk, Sagemaker), GCP, Azure, MLflow, Dagshub, DVC, Grafana, Groq, Nvidia NIM, LangChain, LangGraph, MCP, CrewAI, FAISS, ChromaDB, Neo4j, Redis, PostgreSQL

PROJECTS

Track Your Finance - AI Finance Assistant | FastAPI, React, MongoDB, Gemini, OAuth, Web Scraping

Mar 2026 - May 2026

- Improved personal spending analysis and financial decision support by building an AI-powered finance assistant that categorizes transactions, generates spending insights, and recommends cost-saving credit card options.
- Enabled secure, end-to-end financial analytics workflows by developing a full-stack platform with FastAPI, React, MongoDB, and OAuth, integrating Gemini-based expense analysis with real-time financial data ingestion through web scraping.
- Increased recommendation accuracy and spending categorization reliability by processing and evaluating 10,000+ simulated transactions, validating personalized savings insights and credit card optimization logic.

BonVoyage - AI Travel Webapp | FastAPI, React, PostgreSQL, Redis, AWS EC2/S3/RDS, Llama-3.1 – 8B,

HuggingFace, RapidAPI

Dec 2025 - Feb 2025

- Built an AI-driven travel planning platform** that generated personalized itineraries from natural language prompts, streamlining trip planning and recommendation workflows.
- Developed a multi-stage LLM agent pipeline** using Llama-3.1 via HuggingFace and integrated RapidAPI-based flight and hotel data retrieval to deliver real-time travel suggestions.
- Deployed a scalable full-stack application on AWS** using FastAPI on EC2, React on S3, PostgreSQL on RDS, and Redis caching to improve responsiveness and backend performance.

Interpretable Text-to-SQL Query Generation | Neo4j, FAISS, LLaMA-3.1-8B (Groq), Flan-T5, MySQL

Oct 2025 - Dec 2025

- Improved Text-to-SQL execution accuracy by up to 2.3×** using an interpretable RAG pipeline with **TRC-based logical planning**.
- Integrated Graph RAG** with Neo4j and FAISS for schema-aware context grounding before SQL generation.
- Recovered ~15% failed queries** using a self-healing TRC-to-SQL compiler with execution-error feedback on Spider and SPaC datasets.

Personal RAG Assistant | NVIDIA NIM, LLaMA-3.1-70B, FAISS/Chroma, Pydantic, LangChain, Streamlit

Jul 2025 - Aug 2025

- Improved retrieval performance of a document-level RAG system by incorporating query expansion, re-ranking, and semantic chunking, using MRR/nDCG-based evaluation with Pydantic-structured pipelines.
- Enabled production-ready RAG workflows by implementing MD5-based incremental re-indexing, LangChain orchestration, and a Streamlit interface, ensuring semantic retrieval with NVIDIA Embeddings.

PUBLICATIONS

- K. Sivasubramanian**, R. Charan, S. Narayanan, and L. J. Anbarasi, "Enhanced Plant Disease Classification with Attention-Based Convolutional Neural Networks Using Squeeze-and-Excitation Mechanism," *Frontiers in Artificial Intelligence*, vol. 8, 2025.
- K. Sivasubramanian**, V. Sathiyadevi, M. S. B. M., and M. K. Rajagopal, "Leveraging Deep Learning for Improving Real-Time Stolen Vehicle Tracking in Smart Cities," in *Proceedings of the 11th International Conference on Information Technology: IoT and Smart City*, 2023.